# Project: Network Service

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# 1. Objective

Work as a group of 2--3 students, and build a network service of your choice. Some recommended network services are listed below. If you want to develop some other services, make a case to the instructor, and you can proceed with approval. Extra credit may be granted depending on levels of challenges.

# 2. Network Services

- DNS servers
- E-mail system: SMTP , POP3
- Video streaming
- Voice over IP
- Network Routing Protocol
- This can be very challenging but you can also pick any research paper related to computer networks from a well reputed conference (for example ACM SIGCOMM) and implement thier proposed algorithm or something close to it. For example DCTCP, QUIC

# 3. Instructions

1. **Programming Language:**
    a. You must use **Python** to implement your code (Useful link: https://www.python.org)

   b. Follow the PEP 8 coding guidelines for writing clean and professional Python code. (Here is a link: https://peps.python.org/pep-0008/)
   c. Use `git` for version control of your code
2. **Code Documentation and Comments:**
   a. Properly comment your code to make it clear and understandable
   b. Generate a PDF documentation for your code using a tool like Sphinx that automatically generates it for you. (see this link: https://www.sphinx-doc.org/en/master/). There are other tools that provide similar functionality (i.e. automatically generates documentation for you). You can use any of them if you dont like sphinx
3. **What to Submit:** There should be just one submission per group. All files must be submitted in a **single zip file** . Name the zip file <firstname>_<lastname>_project.zip (replace <firstname> and <lastname> with your name). The zip will have **atleast the following files** :
   a. **Python code** : Python files containing the code you wrote
   b. **Requirements File** : Submit a `requirements.txt` file listing all the Python dependencies required to run your code. (See this link: https://www.geeksforgeeks.org/how-to-create-requirements-txt-file-in-python/)
   c. **Readme** : A `README.md` file that clearly explains: 1- How to compile and run your code. 2- Examples of command-line usage
   d. **Report (PDF)** : The report must be in pdf format. Make sure to include names and rit emails for each student in your group
   e. **Code Documentation** : It must be in PDF format. Use a tool like Sphinx to automatically generate documentation. You can use any other similar tool if you want.
   f. **GIT Log** : Use Git for version control and make commits at appropriate points during development. Submit a git log file showing the commit history. This file should be name `revisions.txt`